CHAPTER 3 km — requested variables that are not directly available

Date: 2026-09-29 (first issued 2026-09-17) · **Scope:** the variable list requested for the CHAPTER regional reanalysis (atmosphere, static, land-surface, ocean waves, sea ice, surface ocean). This note answers one question: of the variables you asked for, which ones can we actually put in the GRIB archive, which ones we can obtain by some other route, and which ones are simply not in the data. It is meant to be read before we commit to a re-conversion of the whole archive, because that re-conversion costs about eleven days of data transfer and has to be done only once. Everything below was verified against the actual files, not against documentation: model configuration read from the wrfout global attributes, and every field checked for real values on independent timesteps in summer 2019, summer 2024 (00/06/12/13/14/18/23 UTC) and March 2024, so that fields which exist in the file but are identically zero are not mistaken for usable data. Accumulated fields were additionally checked for monotonicity within a run, and every ECMWF parameter identifier was round-tripped through the GRIB encoder to confirm name and units. Companion documents: `CHAPTER_VARIABLES.md` is the reference for what the archive does contain, and `CHAPTER_KNOWN_PROPERTIES.md` collects the measured properties and biases of the variables we *do* provide — including the numbers behind several of the caveats in §5 below.

1. Summary

requested block	verdict
Atmosphere, surface (24 variables + wind shear)	**22 of 24 delivered**, plus wind shear. Only the two gust components are missing, and they are excluded by decision (see §4)
Atmosphere, pressure levels (9 variables × 13 levels)	**all delivered**, plus both vertical-velocity forms (`w` in Pa/s and `wz` in m/s) and three extra ERA5 fields: cloud fraction `cc`, rain `crwc` and snow `cswc` water content
TKE at 3 height levels	**not available** — the model does not compute it
Static: orography	**delivered**
Static: land-use / land-cover	**delivered as fractions**: the static file `geo_em.d02` was recovered on 2026-09-18, and its fractional land use gives `cvl`, `cvh`, `tvl`, `tvh`, `slt`, `cl` and `dl` (see §3, §4)
Land-surface	**delivered** as the subset that exists, with ERA5 names — now including the complete radiation budget, runoff, snow cover, snow temperature, albedo, surface stress and the high/low vegetation split
Ocean waves (SWH, period, direction, wave-dependent Cd)	**not available** — no wave model
Sea ice	**concentration delivered**; thickness, snow on ice, ice temperature and albedo not available
Surface ocean (SSH, currents, SST)	**SST delivered**; SSH and currents not available

2. Why: how the archive is produced

The archive is produced from hourly WRF 4.1.1 output (domain d02, Mercator, 1641 × 1353 points at 3 km, 49 vertical levels). The relevant configuration is identical across all periods we

checked (2019, March 2024, July 2024):

option	value	consequence for this list
BL_PBL_PHYSICS	1 (YSU)	non-local scheme: **no prognostic TKE**. PBLH was also not written out
KM_OPT	4	no prognostic turbulent kinetic energy array either
SF_OCEAN_PHYSICS	0	**the ocean is not simulated**: no currents, no sea surface height, no mixed layer
(no wave model)	—	**no wave fields** of any kind, and therefore no wave-dependent drag coefficient
SST_UPDATE	0	SST is **constant within each 24 h run**, re-initialised at every daily run
CU_PHYSICS	0	convection-permitting: RAINC is identically zero, so total precipitation is complete in RAINNC
MP_PHYSICS	6 (WSM6)	cloud, rain, ice, snow and graupel are available; **no hail category**
ICLOUD	1 (Xu-Randall)	cloud fraction is **continuous**, not a 0/1 flag — see §5
SF_SURFACE_PHYSICS	3 (RUC)	6 soil levels at 0, 5, 20, 40, 160, 300 cm (ERA5 uses 4 layers)
SF_LAKE_PHYSICS, SF_URBAN_PHYSICS	0, 0	no lake model, no urban canopy

A separate point worth making explicitly: some fields are present in the files but are **identically zero on every timestep we sampled**, and are therefore not usable. These are ACHFX, ACLHF, ACGRDFLX, NOAHRES, SSTSK, SST_INPUT, SWNORM, ACSNOW, HAILNC, RAINC, RAINSH, PREC_ACC_C, and the downward longwave at the top of the atmosphere. None of them will be written to the GRIB files.

3. Variables we can obtain by another route

These are not in the model output, but there is a way to approximate them. None of them is implemented, and none would be published under an ERA5 name: each is our own diagnostic, not the model's. Items that appeared here in previous versions have moved: **surface albedo is now delivered** (al, the snow-free background, and fal, the actual all-sky ratio), and **fractional land cover is now delivered too** — the static file geo_em.d02, recorded here as irrecoverable, was obtained from LRZ on 2026-09-18 and is the run's own (its land use agrees with the model's dominant category on 100.00 % of points once the lake recode is excluded). It brought cvl, cvh, tvl, tvh, slt, cl and dl.

variable	route	confidence	cost
Sea-ice surface temperature	Use the skin temperature we already derive, restricted to points where sea-ice concentration > 0	**Medium.** It is a radiative skin temperature, consistent with the model's own longwave emission, but not a dedicated ice-surface temperature	Free
Drag coefficient **without** wave effect	From the model's own roughness length and stability: WRF's surface layer computes `Cd = (κ / ψ10)²` with `ψ10 = ln((10 + z0)/z0) − ψm(10 m)` (`module_sf_sfclayrev.F`, v4.1.1); `(κ / ln(10/z0))²` is only its neutral limit	**High** if the stability function is recomputed from the archived fields; the neutral formula alone is not the drag the simulation used	Free
Planetary boundary layer height	Bulk Richardson number on the virtual potential temperature profile; all required fields are present	**Low as an ERA5 field** — it would be our diagnosis, not the model's: YSU's own `PBLH` was not written, so this fails the publication rule and is not delivered as `blh`	Free

4. Variables we cannot provide

variable	why	the only way to get it
`10efg`, `10nfg` (eastward / northward gust components)	Only a gust *magnitude* proxy exists (see §5), and it is the maximum of the resolved wind, not a gust scheme. Their direction would have to be assumed equal to the mean wind, at an instant that does not even coincide with the maximum	Decomposing the magnitude along the 10 m wind is possible and cheap, but it was judged too weak to publish under ERA5 names, so **it was decided not to produce them**
TKE (at any level)	The PBL scheme is YSU, which is non-local and carries no turbulent kinetic energy; the diffusion option does not produce one either. The field does not exist, at any height	Re-run WRF with a TKE-carrying PBL scheme (MYNN or MYJ). A similarity-theory estimate would need boundary-layer height *and* surface heat flux, both themselves missing, so the errors compound: it would look plausible but would not be consistent with the simulation

variable	why	the only way to get it
Significant wave height, mean wave period, wave direction	No wave model was coupled to the atmosphere	A wave model run, or an external wave reanalysis (ERA5 waves at 0.36°, ~40 km) — which loses exactly the coastal detail that motivates a 3 km product
Drag coefficient including wave effect	Same reason: the surface exchange used Charnock roughness with no sea state	As above. The wave-free drag coefficient *is* available (§3)
Sea surface height, ocean currents (u, v)	The ocean is not simulated	An ocean reanalysis (e.g. CMEMS), regridded
Sea-ice thickness, snow thickness on sea ice	Not simulated. Sea ice enters only as a concentration boundary field	An external sea-ice product
Sea-ice albedo as a prognostic ice property	Not simulated	Approximable radiatively (§3), but not as an ice-model variable
Snowmelt	The field exists (`ACSNOM`) but is **not a monotone accumulator**: individual points drop back to zero within a single run (measured -7.26 mm against 00 UTC on 2024-07-01), so the difference of two consecutive hours is not the melt of that hour, as it is for the other accumulations. Published under the ERA5 name it would be wrong	Would need the melt diagnosed per timestep inside the model
Surface sensible and latent heat flux	**The model computed them and the output stream does not carry them.** `isfflx = 1`, so they were produced at every timestep; they were simply not written to the history stream, and no untrimmed output exists anywhere. The accumulated `ACHFX`/`ACLHF` are identically zero because they are **Noah** accumulators in a RUC run, not because of a bucket setting	Only a re-run writing them. Reconstruction from the surface energy budget fails: the ground heat flux is missing too, leaving two unknowns in one equation
Skin temperature as a model variable	`TSK` was not written out — a deliberate trimming of the output stream, not an accident	We already provide `skt`, obtained by inverting the upward longwave flux (see §5), which over water recovers the model's own SST to 0.0005 K

variable	why	the only way to get it
Maximum and minimum 2 m temperature (`mx2t`, `mn2t`)	`output_diagnostics = 0`, so WRF never wrote `T2MAX`/`T2MIN`. An hourly sample of `T2` is **not** the model's within-hour extremum, so no work on the converter produces these	Only a re-run with `output_diagnostics = 1`. Note `nwp_diagnostics = 1` **is** on, which is what gives `WSPD10MAX` and six other per-hour maxima — the run carries within-hour extrema, just not for temperature
Any pressure level above 50 hPa (10 hPa, and ERA5's other seven: 1, 2, 3, 5, 7, 20, 30 hPa)	The model top is at 50 hPa (`p_top_requested = 5000` Pa in the run's namelist) and the highest model level sits at 51.85–51.96 hPa (measured, 2024-04-15 12 UTC). Above it the run holds no atmosphere at all: a 10 hPa field would be an extrapolation 40 hPa past the lid, not a model value	Only a re-run with the top near 5 hPa — more vertical levels, and the damping layer this run had switched off (`damp_opt = 0`). ERA5's 10 hPa is not a substitute under a CHAPTER name
Planetary boundary layer height (`blh`)	YSU **does** diagnose one; it was not written to the history stream. The limit is the output, not the scheme	Only a re-run writing `PBLH`. The bulk Richardson diagnostic of §3 would be ours, not the model's, so it is not published as `blh`

Dropped on 2026-09-17 and **reinstated on 2026-09-18**, once the objection had an answer:

Variable	Why it was dropped	Why it is now published
Soil moisture and soil temperature (`swvl1-4`, `stl1-4`)	The scheme carries point values at 0, 5, 20, 40, 160 and 300 cm; ERA5's four layers are averages over 0-7, 7-28, 28-100 and 100-289 cm	The objection was against *relabelling* the point values, and it stands. But they can be **integrated**: the scheme is a level scheme with a linear profile between nodes, and ERA5's whole column lies inside those nodes (289 cm is above the deepest, 0 cm is the first), so each layer is the exact integral and nothing is extrapolated
Vegetation and soil type (`tvl`, `tvh`, `slt`)	The categories are WRF's MODIS-IGBP and STATSGO ones, while those paramIds carry ECMWF's own code tables	`slt` is no longer a translation at all: ECMWF's seven types are *defined* by clay/sand thresholds, and the recovered static file carries clay and sand fractions, so the definition is applied directly. `tvl`/`tvh` do use a table, published in full in `CHAPTER_VARIABLES.pdf` §8; every row is a semantic identity, and the one question MODIS cannot answer (evergreen vs deciduous shrubs) is left **missing** rather than guessed

Variable	Why it was dropped	Why it is now published
Vegetation cover (`cvl`, `cvh`)	They rested on one dominant category per cell, while ERA5 lets low and high vegetation coexist	The recovered static file has the fractional cover, so they are simple sums over the low and high classes — the same quantity ERA5 means. This was the right objection: 17.9 % of land cells carry both above 5 %, and the dominant category holds only 90.1 % of a cell on average

Still not published:

Variable	Why
`lai_lv`, `lai_hv`	The model carries a single LAI per cell; ERA5 wants one per vegetation class, defined per unit of vegetated area. Any split between the two would be our assumption. The model's own LAI can be delivered under its own name if wanted
`anor`, `isor`	The static file carries the Kim-Arakawa asymmetry and effective-length parameters. ECMWF's angle and anisotropy come from the eigenstructure of the terrain-gradient covariance: related, not the same, and the difference does not fit in one line

If either is needed, say so: the underlying WRF fields exist and can be delivered under their own names (`LAI`, `VEGFRA`, `OA1-4`, `OL1-4`), where nothing is claimed about ERA5 equivalence.

5. Caveats on the variables we *do* provide

These matter for correct use and should be read together with the variable list.

- **Cloud cover is fractional.** An earlier draft of this note said the model's cloud fraction was binary; that was wrong, and the correction is in our favour. The scheme is Xu-Randall and the field is genuinely continuous: over a 500 × 600 sub-domain only 18-22 % of cloudy points sit exactly at 1, the rest spread across the whole 0-1 range. Total, high, medium and low cloud cover are maximum-random overlaps of that field, and the cloud fraction is also delivered per pressure level as `cc`.
- **SST is constant within each day.** Each daily run is initialised with a fixed sea surface temperature and does not update it. The field therefore has a daily, not hourly, time resolution.
- **Skin temperature is derived, not modelled.** `TSK` was not written out, so it is obtained by inverting the upward longwave flux, `LWUPB = eps sigma T^4 + (1-eps) LWDN`, with the emissivity of the dominant land-use category (`VEGPARM.TBL`, `MODI-RUC` section, 0.85 barren to 0.98 water), measured from the run itself. Over open water the model's skin temperature is the SST and the inversion recovers it to 0.0005 K RMSE; on snow-free land it matches the 0 cm soil level to 0.026 K RMSE (one September timestep, 2024-09-04 16Z). Snow switches the emissivity to 0.98 above a snow-cover threshold, and that threshold is the one free parameter of the reconstruction: land `skt` RMSE is 0.0012 K in July and 0.084-0.119 K in January and February. The figures are in `CHAPTER_KNOWN_PROPERTIES.md` §2.
- **The slope of orography is the resolved one.** ERA5's `slor` is a sub-grid parameter of the form-drag scheme; what we write is the slope of the resolved 3 km terrain. (`sdor` is instead a genuine sub-grid standard deviation, computed by the pre-processor.)

- **Water contents are specific, not mixing ratios.** WRF carries mixing ratios, per kg of dry air; `clwc`, `ciwc`, `crwc`, `cswc` and the column integrals are converted to specific contents, per kg of moist air, as their ECMWF definitions require. The correction is 0.011 % in the median, 1.44 % at the 99th percentile and 2.26 % at most, measured on the full 3-D field; the median is small because most of a column is dry upper troposphere.
- **Convective inhibition is not defined everywhere.** By construction it is only computed where CAPE reaches 100 J/kg. `mucin` is **missing** (a GRIB bitmap) everywhere else: 94.3 % of the domain on 2024-02-15 12Z, 71.2 % on 2024-07-15 12Z. Zero would be wrong there, because it would read as "no inhibition" exactly where the inhibition is largest. `mucape` keeps a **zero** where the routine returns nothing, which is the correct reading for CAPE: it is zero on 69 % of the domain in the February case and 46 % in the July one.
- **CAPE and CIN are most-unstable**, computed from the 500 m deep parcel with the highest equivalent potential temperature in the lowest 3 km — not surface-based.
- **The gust field is not a gust parameterisation, and the distance is large enough to state.** It is the maximum of the *resolved* 10 m wind speed over the output hour — the best proxy the run can give, but ERA5's `10fg` is a parameterised gust, roughly 1.6 times its own mean wind. Measured over the audit sample: ours is **0.625 of ERA5's `10fg`** and 0.644 of its instantaneous `i10fg`, with a correlation of 0.81, while against **our own** instantaneous 10 m wind in the same file it sits at **1.004 to 1.032** — so the hourly maximum adds half a per cent to three per cent and nothing more. Reading it as a gust under-predicts by about 38 %, and that is a difference of quantity, not a bias to correct. Its declared one-hour window is also right only 88 % of the time. Both are stated in full, with the numbers, in `CHAPTER_KNOWN_PROPERTIES.md` §5.
- **The land/sea mask changes with the season.** Where sea ice forms, the model reclassifies the point as ice: the land mask, the vegetation and soil types and the background albedo all change there (2469 points on 20 March 2024, exactly the sea-ice points). The background albedo also changes for a second reason: it is a two-season table value, and it moves on 89.9 % of the land points at the boundary between the winter (November-March) and summer (April-October) tables. `lsm` and `al` are therefore written in every hourly file and must not be treated as static. `tvl`, `tvh` and `slt` do not move: they come from the static file, not from the hourly mask, and the soil columns `swvl1-4`/`stl1-4` are masked on the permanent land, so sea ice carries no soil column.
- **Surface stress is a diagnostic, not a model output.** The eastward and northward stresses are reconstructed as density × friction velocity squared, aligned with the 10 m wind. The friction velocity is the model's own, so this is the same similarity theory the simulation used, but the alignment with the mean wind is an assumption.
- **Snow temperature only exists where there is snow.** The field is the temperature at the top of the snow/soil column; away from snow it is simply a soil temperature, so it is written as missing there rather than published as a snow temperature that does not exist.
- **Albedo comes in two forms.** `al` is the snow-free climatological background the model was initialised with; `fal` is the actual ratio of upward to downward shortwave, which is what the simulation really used — but it is undefined at night and is written as missing below 50 W/m² of incoming shortwave.
- **Three runoff variables carry one field's worth of information.** The sub-surface component is essentially empty at daily scale: measured over the audit sample, `ssro` is **identically zero on five of the eighteen **sampled hours and, where it is not, non-zero at** at most 19 points of 2 220 273**, with a domain mean four to eight orders of magnitude below `sro`. So **`ro` equals `sro` to four significant figures at every timestep** — the identity `ro = sro + ssro` holds to 3.7e-9 m, there is simply nothing in the second term. The cause is the time reference, not the conversion: the model's drainage counter is alive, but it barely moves *within a day*, and `ssro` is the drainage of one hour. Against ERA5 the partition is the other way round (`sro` 1.0–4.6× ERA5, `ssro` 0.000–0.004×, `ro` 0.19–0.85×), so **do not close a water budget assuming this archive carries a drainage term.** All three are published because `ro` is the ERA5-canonical total; `ro` and `sro` are knowingly duplicate messages.

- **There is no graupel field, and `tcw` carries it anyway.** ECMWF defines `tcw` as the sum of five species — vapour, liquid, ice, rain, snow — and ours is the sum of **six**, graupel included. ECMWF does define `260028 grle` and `260001 tcolg`, but ERA5 publishes neither, so there is nowhere to put the graupel except inside `tcw`. **Reconstructing `tcw` from the five published component columns therefore does not reproduce it**, and the difference is the graupel: 0.4 % of the condensate in the domain mean, invisible — but **up to 26.5 kg/m² inside a July convective cell**, which is more than the rest of the column's condensate put together.
- **Rainfall excludes snow and graupel, not hail.** The microphysics has no hail category, so rain is total precipitation minus snow minus graupel.
- **2 m dewpoint is now in kelvin.** The previous GRIB1 archive wrote it in degrees Celsius under a parameter that ECMWF defines in kelvin. If you read `2d` from the old files, it needs +273.15.
- **Accumulated fields cover the hour ending at the validity time**, as in ERA5, not the time since the start of the model run. In the GRIB files they carry a one-hour step with the previous hour as reference time, so the validity time is the timestep itself; the 00 UTC file holds the hour 23→00 UTC, taken from the previous day's run. The 10 m wind maximum covers the same hour, because the model resets it at every output.
- **Geolocation is now exact to 2 m** (2.009 m at most, measured). The grid declares the spherical earth of radius 6370 km that WRF actually integrates on. The previous GRIB1 files could not express this and were off by up to ~1.1 km at the northern edge of the domain.

---

6. What we need from you

The questions about the gust components and about TKE that appeared here previously have been settled: neither is produced. The one about the static `geo_em` file is also closed, but the other way round from what this document said last: the file **was** obtained from LRZ on 2026-09-18, and everything that depended on it is now delivered (§3, §4).

1. **The land-surface list.** The original proposal was lost. We deliver, under ERA5 names: soil moisture and soil temperature (**4 layers, as true ERA5 layer averages**), snow depth, density, cover and temperature, snowfall, surface, sub-surface and total runoff, skin reservoir content, **high and low vegetation cover and type**, **soil type**, **lake cover and depth**, friction velocity, surface roughness, the two surface albedos, the two components of surface stress, and the complete radiation budget (downward, net and clear-sky, at the surface and at the top of the atmosphere). Two items of the original list are **not** among them: **leaf area index**, because the model carries one value per cell and ERA5 wants one per vegetation class (§4), and **snowmelt**, because the field exists but is not a monotone accumulator (§4). Please confirm or amend.
2. **Wind shear: which definition?** We currently deliver the bulk 10 m to 100 m shear. Any other definition (fixed layers, per pressure level, 0-6 km bulk) is a one-line change, but it has to be decided before the archive is re-converted.
3. **Vertical levels.** We deliver 13 pressure levels. ERA5 uses 37, of which 29 would be within this model's domain (the model top is at 50 hPa, so nothing above it can be produced). Going to 29 levels would take the hourly file from 0.74 GB (measured mean over September 2024) to ~1.45 GB (a projection made on 2026-09-17), i.e. a year of archive from 6.5 TB to ~12.7 TB; on those grounds we kept 13. Say so now if the denser set matters for your use.